3-17: 205-206页1,5,8,12,13; 208页1,3,5.
3-20: 205页4; 208页2,4; 219页4(c); 225-226页4,10.
Denote by p_A the minimal polynomial of A, and f_T the characteristic polynomial of A.

205-1

Let $T \in \mathcal{L}(\mathbb{R}^2)$ such that its matrix is

$$A = \begin{pmatrix} 1 & -1 \\ 2 & 2 \end{pmatrix}$$

(a) Prove that the only subspace of $\mathbb{R}^2$ invariant under T are $\mathbb{R}^2$ and 0.
(b) If $U \in \mathcal{L}(\mathbb{C}^2)$ with matrix A, U has a 1-dimensional invariant subspace.
Proof: (a) Note that $f_A(x) = (x-1)(x-2)+2 = x^2-3x+4$ so f_A has no roots in $\mathbb{R}$, hence no eigenvalues and no 1-dimensional invariant subspaces.
(b) f_A has roots in $\mathbb{C}$, hence a 1-dimensional invariant subspace.

205-5

Every matrix A such that $A^2 = A$ is similar to a diagonal matrix.
Proof: Since $A^2 = A$, $p_A \mid x(x-1)$ has only simple zeros, so A is diagonalizable.

205-8

Let $T \in \mathcal{L}(V)$. If every subspace of V is invariant under T, then T is a scalar multiple of the identity operator.
Proof: Since every 1-dimensional subspace is $T-$invariant, every (nonzero) vector is an eigenvector of T.
If eigenvectors v, w correspond to different eigenvalues λ, μ, then $T(v+w) = \lambda v + \mu w \notin \operatorname{Span}(v+w)$, a contradiction. Hence they have a common eigenvalue c, and $T = cI$.

206-12

Let $T \in \mathcal{L}(V)$ where V is a vector space over F which is algebraically closed. Let $f \in F[x]$. Prove that c is an eigenvalue of $f(T)$ iff $c = f(t)$, where t is an eigenvalue of T.
Proof: If t is an eigenvalue of T, suppose $Tv = tv$ for $v \neq 0$, then $f(T)v = f(t)v = cv$ so c is an eigenvalue of $f(T)$.
If c is an eigenvalue of $f(T)$, the polynomial $f(x) - c$ splits in $F[x]$, so $f(x) - c = a(x-\lambda_1)\cdots(x-\lambda_k)$. Then $f(T) - cI = a(T-\lambda_1 I)\cdots(T-\lambda_k I)$. Since $\det(f(T)-cI) = 0$, $\exists 1 \leqslant i \leqslant k, \det(T-\lambda_i I) = 0$, so λ_i is an eigenvalue of T where $c = f(\lambda_i)$.

206-13

Let $V = F^{n\times n}$, and $A \in F$ be fixed. Let $T, U \in \mathcal{L}(V)$, defined by $T(B) = AB$, and $U(B) = AB - BA$.
(a) True or false? If A is diagonalizable, then T is diagonalizable.
(b) True or false? If A is diagonalizable, then U is diagonalizable.
Solution: (a) True. In Exercise10 of page 198 we showed that $p_A = p_T$ so A is diagonalizable iff T is diagonalizable.

(b) True. Consider $S \in \mathcal{L}(V), S(B) = BA$, then $ST = TS$, and if A is diagonalizable, then S, T are both diagonalizable. Hence they are simultaneously diagonalizable, and $U = T - S$ is also diagonalizable.

208-1

Find an invertible real matrix P such that $P^{-1}AP$ and $P^{-1}BP$ are both diagonal, where A, B are the real matrices:

$$(a) : A = \begin{pmatrix} 1 & 2 \\ 0 & 2 \end{pmatrix}, B = \begin{pmatrix} 3 & -8 \\ 0 & -1 \end{pmatrix}$$
$$(b) : A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, B = \begin{pmatrix} 1 & a \\ a & 1 \end{pmatrix}$$

Solution:(a) Let $P = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$, then $P^{-1}AP = \mathrm{diag}(1, 2)$ and $P^{-1}BP = \mathrm{diag}(3, -1)$.
(b) Let $P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, then $P^{-1}AP = \mathrm{diag}(2, 0)$ and $P^{-1}BP = \mathrm{diag}(1 + a, 1 - a)$.

208-3

Let T be a linear operator on an $n-$dimensional space, and suppose that T has n distinct eigenvalues. Prove that any linear operator which commutes with T is a polynomial in T.
Proof: Since T has n distinct eigenvalues, T is diagonalizable. Take a basis $\mathcal{B}$ such that $[T]_{\mathcal{B}} = \mathrm{diag}(c_1, \cdots, c_n)$, then if U commutes with T, U is also diagonal. Suppose $U = \mathrm{diag}(d_1, \cdots, d_n)$. Then by Lagrange's interpolation, $U = f(T)$ (i.e., $d_i = f(c_i)$) for some polynomial $f \in F[x]$.

208-5

Let $V = F^{n \times n}$. If $A \in V$, let $T_A \in \mathcal{L}(V), T_A(B) = AB - BA$. Consider the family of linear operators T_A obtained by letting A vary over all diagonal matrices. Prove that the operators in that family are simultaneously diagonalizable.
Proof: If A, B are both diagonal, then $AB = BA$ implies

$$T_A T_B(C) = T_A(BC - CB) = ABC - ACB - BCA + CBA = T_B(AC - CA) = T_B T_A(C)$$

so these operators T_A are pair-wise commutative.
Recall Exercise13 in Page206, if A is diagonal then T_A is diagonalizable. Hence they are simultaneously diagonalizable.

205-4

Let

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 2 & -2 & 2 \\ 2 & -3 & 2 \end{pmatrix}.$$

Is A similar over $\mathbb{R}$ to a triangular matrix? If so, find such a triangular matrix.

Solution: $f_A(x) = x^3$ so $p_A \mid f_A$ splits in $\mathbb{R}$. Hence A is upper triangular. Let $P = \begin{pmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ -2 & 2 & 0 \end{pmatrix}$, then

$$P^{-1}AP = J = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

208-2

Let $\mathcal{F}$ be a commuting family of 3×3 complex matrices. How many linearly independent matrices can $\mathcal{F}$ contain? What about the $n \times n$ case?

Solution: We show that the answer is $g(n) = \lfloor n^2/4 \rfloor + 1$:

Consider $A = \begin{pmatrix} 0 & M \\ 0 & 0 \end{pmatrix}$ where $M \in F^{k \times n-k}$, then $B = \begin{pmatrix} 0 & N \\ 0 & 0 \end{pmatrix}$ satisfy $AB = 0 = BA$.

Hence take $k(n-k)$ independent elements in the form of A, and $I_{n \times n}$, then they are commuting independent matrices. Let $k = \lfloor n/2 \rfloor$ then $k(n-k) + 1 = \lfloor n^2/4 \rfloor + 1$.

Now we prove that $|\mathcal{F}| \leqslant g(n)$: Note that if $d_1 + \cdots + d_k = n$, then $g(n) \geqslant g(d_1) + \cdots + g(d_k)$. (Use induction on k).

Suppose $\mathcal{F}$ is a maximal class of commuting matrices $F^{n \times n}$, where F is a field. We can assume that F is algebraically closed, by taking closure. We prove by induction that $|\mathcal{F}| \leqslant g(n)$: the base $n = 1$ is trivial.

If $A \in \mathcal{F}$ has at least two distinct eigenvalues, suppose $p_A(x) = (x - \lambda_1)^{r_1} \cdots (x - \lambda_k)^{r_k}$, and $W_i = \mathrm{Ker}(\lambda_i I - A)^{r_i}$, then $W = W_1 \oplus \cdots \oplus W_k$ and W_k is invariant under all $T \in \mathcal{F}$. Hence $\mathcal{F}_i = \{T_i : T \in \mathcal{F}\}$ are commuting matrices in $F^{d_i \times d_i}$ where $d_i = \mathrm{dim} W_i$, so

$$|\mathcal{F}| \leqslant \sum_{i=1}^{k} |\mathcal{F}_i| \leqslant \sum_{i=1}^{k} g(d_i) \leqslant g(n).$$

Otherwise suppose all $A \in \mathcal{F}$ has only one eigenvalue, i.e., there is a $r_a \in K$ such that $A - r_a I$ is nilpotent. Let $R = \langle \mathcal{F} \rangle \subset \mathcal{L}(V)$, then since R is maximal $I \in R$, so R is an $F-$algebra. Every $T \in R$ can be written as $cI + \sum c_i(A - r_a I)$, where $A - r_a I$ are nilpotent, hence T is nilpotent if $c = 0$ and a unit if $c \neq 0$, so R has a unique prime(maximal) ideal M (the nilradical & Jacobsan radical). Since $R = KI \oplus M$, $R/M \cong K$ so $\mathrm{dim} R = \mathrm{dim} M + 1$.

Note that V is a (faithful) $R-$module, so V/MV is a $R/M \cong K$ vector space. Let $k = \mathrm{dim} V/MV$ and take a basis $\{\alpha_1 + MV, \cdots, \alpha_k + MV\}$, then by Nakayama's lemma $\alpha_1, \cdots, \alpha_k$ generate V as an $R-$module. Consider $\varphi : M \to M\alpha_1 \oplus \cdots \oplus M\alpha_k, f \mapsto (f\alpha_1, \cdots, f\alpha_k)$, then V is faithful so φ is injective, hence $\mathrm{dim}_K M \leqslant \sum \mathrm{dim}_K M\alpha_i \leqslant k \mathrm{dim}_K MV \leqslant k(n-k)$. Therefore $\mathrm{dim} R \leqslant 1 + k(n-k) \leqslant g(n)$.

208-4

Let $A, B, C, D \in \mathbb{C}^{n \times n}$ commute with each other. Let E be the $2n \times 2n$ matrix

$$E = \begin{pmatrix} A & B \\ C & D \end{pmatrix}.$$

Prove that $\det E = \det(AD - BC)$.
Proof: Since A, B, C, D commute with each other, we can take $P \in \mathbb{C}^{n\times n}$ such that $P^{-1}AP, P^{-1}BP, P^{-1}CP, P^{-1}DP$ are all upper triangular. Then

$$\det E = \det \begin{pmatrix} P^{-1} & \\ & P^{-1} \end{pmatrix} \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} P & \\ & P \end{pmatrix} = \det \begin{pmatrix} A' & B' \\ C' & D' \end{pmatrix}$$

Suppose $A' = (a_{ij})$, then $\prod_{i=1}^{n} (a_{ii}d_{ii} - c_{ii}b_{ii}) = \det(A'D' - B'C') = \det(AD - BC)$. Let $M_{ij} = \begin{pmatrix} a_{ij} & b_{ij} \\ c_{ij} & d_{ij} \end{pmatrix}$, then $M = (M_{ij})_{1\leqslant i,j\leqslant n}$ can be transformed into $\begin{pmatrix} A' & B' \\ C' & D' \end{pmatrix}$ by exchanging rows and columns. Hence

$$\det E = \det M = \prod_{i=1}^{n} \det M_i = \det(AD - BC).$$

219-4(c)

Let $T \in \mathcal{L}(V)$. Suppose $V = W_1 \oplus \cdots \oplus W_k$, where each W_i is $T-$invariant. Let T_i be the induced operator on W_i.
Prove that $p_T = \operatorname{lcm}(p_{T_1}, \cdots, p_{T_k})$.
Proof: Clearly $p_T(T_i) = 0$ so $p_{T_i} \mid p_T$, hence $\operatorname{lcm}(p_{T_1}, \cdots, p_{T_k}) \mid p_T$.
Since $\operatorname{lcm}(p_{T_1}, \cdots, p_{T_k})(T_i) = 0 \forall 1 \leqslant i \leqslant k$, $\operatorname{lcm}(p_{T_1}, \cdots, p_{T_k})(T) = 0$ so $p_T = \operatorname{lcm}(p_{T_1}, \cdots, p_{T_k})$.

226-5

Let V be a finite dimensional vector space over F, and $T \in \mathcal{L}(V)$. Define $T \in \mathcal{L}(V)$ by $T(B) = AB - BA$.
Prove that if A is nilpotent, then T is nilpotent.
Proof: If $A^n = 0$, then $U : B \to AB$ is nilpotent, and $V : B \to BA$ is nilpotent. Note that $UV = VU$, so $T = U - V$ is nilpotent since $T^{2n} = \sum_{k=0}^{2n} U^k(-V)^{2n-k}\binom{2n}{k}$.

226-10

Let T be a linear operator on the finite dimensional space V, let $p = p_1^{r_1} \cdots p_k^{r_k}$ be the minimal polynomial for T, and let $V = W_1 \oplus \cdots \oplus W_k$ be the primary decomposition for T, i.e., $W_i = \operatorname{Ker} p_j(T)^{r_i}$. Let W be any subspace of V which is invariant under T. Prove that

$$W = (W \cap W_1) \oplus \cdots \oplus (W \cap W_k).$$

Proof: Let $f_i = \prod_{j\neq i} p_j^{r_j}$, then $f_i \cdot w_j = 0 \forall w_j \in W_j, j \neq i$.
For each $w \in W$, suppose $w = w_1 + \cdots + w_k$ where $w_i \in W_i$, then $f_i(w) = f_i(w_i) \in W$ implies $f_i(w_i) \in W \cap W_i$. Since f_i and p_i are coprime, there exists u, v such that $uf_i + vp_i^{r_i} = 1$ so $w_i = u(f_i(w_i)) \in W \cap W_i$. Hence $W = (W \cap W_1) + \cdots + (W \cap W_k)$.
If $w_i \in W \cap W_i$ satisfy $\sum_{i=1}^{k} w_i = 0$, then $f_i(w_i) = f_i\left(\sum_{i=1}^{k} w_i\right) = 0$ so $w_i = u(f_i(w_i)) = 0$. Therefore $W = (W \cap W_1) \oplus \cdots \oplus (W \cap W_k)$.